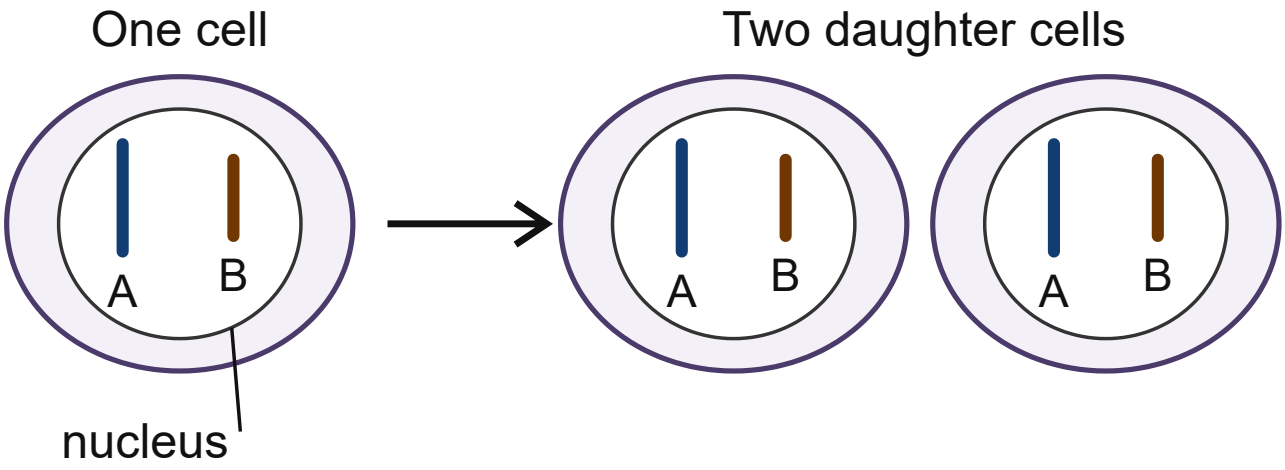
Reference A matching chromosome sets



Compare the full A and B set in both daughter cells. Each new cell has a matching set.

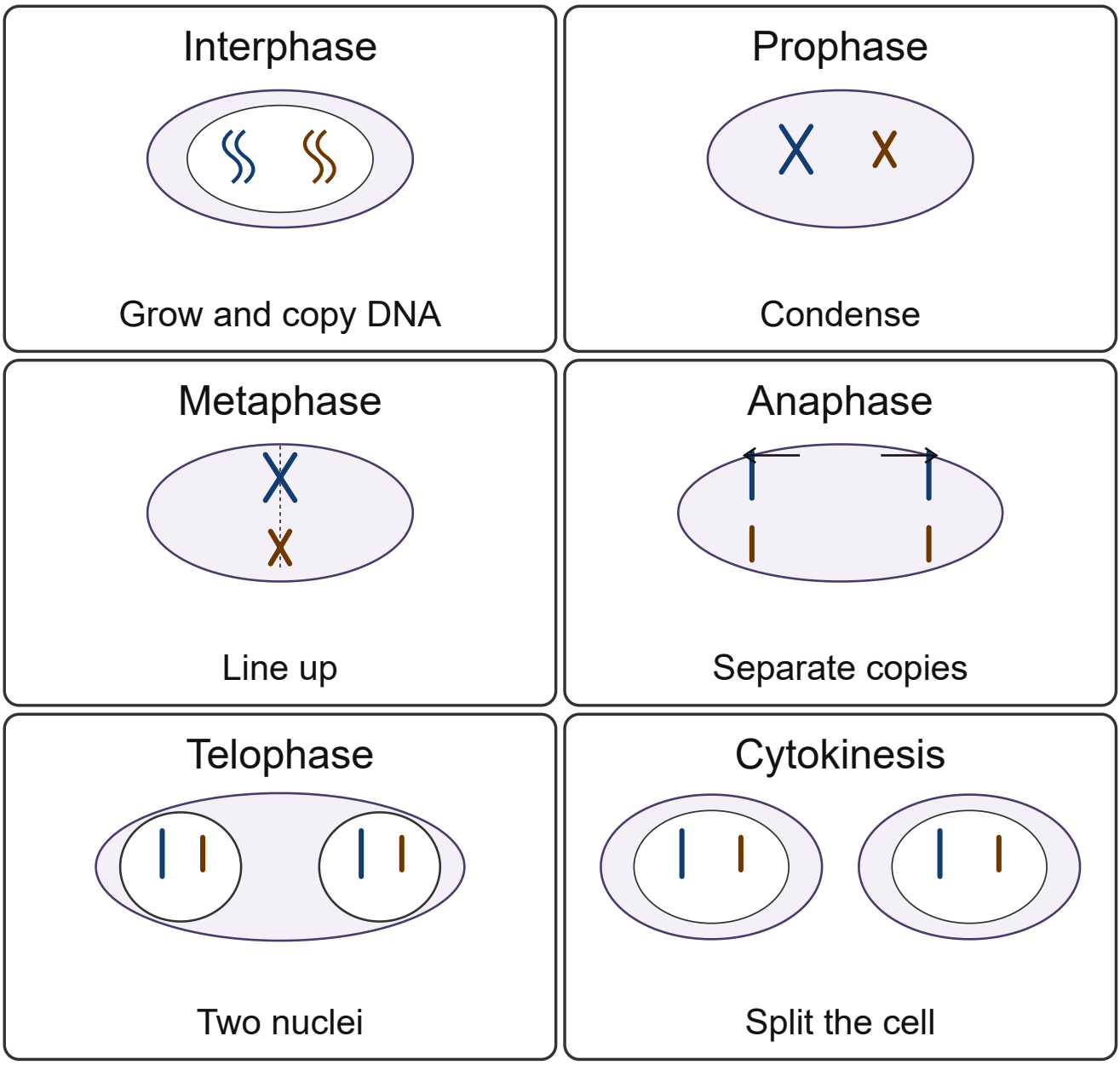
Reference only. Use labels as well as colours. The symbols are simplified chromosome types, not a full picture of real DNA.

Step	What changes
Interphase	The cell grows and copies its DNA.
Mitosis	Copied chromosomes separate into two nuclei.
Cytokinesis	The cell splits into two daughter cells.

In the number case, six chromosomes in the starting body cell means six in each daughter after ordinary mitosis and cell division. A fresh case starting with eight gives eight in each daughter.

Made by Matt

Reference B the six named stages



Interphase, prophase, metaphase, anaphase, telophase and cytokinesis. Four middle phases belong to mitosis.

Reference only. Interphase comes before mitosis. Cytokinesis follows the formation of the two nuclei. The four named mitosis phases are prophase, metaphase, anaphase and telophase.

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Arrival and exit Aspire

Use the labelled picture on Reference A for question 1. Respond in your usual mode.

1. Using the picture alone, state the number of separate cells after the final split.

2. Which labelled cell structure contains the chromosome set?

3. Explain what the A and B labels show about the sets in the two new cells.

4. Give one change shown by the model and one thing its simplified symbols do not show.

Exit ticket

1. Why must DNA copying happen before mitosis?

2. Why does splitting the cell alone not explain matching chromosome sets?

Pupil voice

What helped you most today, or what would you change next time?

Point, speak, sign, draw, write or choose. You may pass.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

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You do Aspire

Use your book or this sheet. Use Reference A and B. One response per task.

1. Predict what would happen to the sets if copied chromosomes did not separate evenly. Give a reason. [3]
Staff I ☐ S ☐ C ☐

---

2. Explain why swapping mitosis and DNA copying would break the model. [3]
Staff I ☐ S ☐ C ☐

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3. Fault finder: compare two faulty models. One skips DNA copying; the other puts both copies of one chromosome in one nucleus. Find each first error and propose a repair. [4]
Staff I ☐ S ☐ C ☐

Feedback and response · Aspire

Feedback codes

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt